INDEX NUMBERS IN R

Example: The data in the table below show the market prices and quantities purchased of four selected items for January 2006 and January 2026.

Item	January 2006		January 2026	
	Price (₦)	Quantity	Price (₦)	Quantity
Toothpaste(9 oz)	120	6	200	6
Shampoo (7 oz)	320	4	350	5
Cough Tablets (pkt of 10)	150	2	175	3
Antiperspirant (2 oz)	300	3	380	4

Using January 2006 as the base year, determine (i) Laspeyres' price index (ii) Paasche's price index (iii) Dorbish and Bowley's price index (iv) Fisher's ideal price index and (v) Marshall-Edgeworth price index. Use hand calculator and R package.

Solution:

```
# Data

item <- c("Toothpaste", "Shampoo", "Cough Tablets",
"Antiperspirant")

p0 <- c(120, 320, 150, 300)   # Prices in 2006

q0 <- c(6, 4, 2, 3)           # Quantities in 2006

p1 <- c(200, 350, 175, 380)   # Prices in 2026

q1 <- c(6, 5, 3, 4)           # Quantities in 2026

# Create data frame

data <- data.frame(item, p0, q0, pn, qn)
```

```
# Display data
print(data)


# Required Summations
sum_p0q0 <- sum(p0 * q0)
sum_p1q0 <- sum(p1 * q0)
sum_p0q1 <- sum(p0 * q1)
sum_p1q1 <- sum(p1 * q1)
sum_p1_q0q1 <- sum(p1 * (q0 + q1))
sum_p0_q0q1 <- sum(p0 * (q0 + q1))
# (i) Laspeyres Price Index
L <- (sum_pnq0 / sum_p0q0) * 100
# (ii) Paasche Price Index
P <- (sum_pnqn / sum_p0qn) * 100
# (iii) Dorbish and Bowley Index
DB <- (L + P) / 2
# (iv) Fisher's Ideal Index
Fisher <- sqrt(L * P)
# (v) Marshall-Edgeworth Index
ME <- (sum(pn(q0+qn) )/ (sum(p0(q0+qn)) * 100
```

Results

```
cat("Laspeyres Price Index =", round(L, 2), "\n")

cat("Paasche Price Index =", round(P, 2), "\n")

cat("Dorbish & Bowley Price Index =", round(DB, 2), "\n")

cat("Fisher's Ideal Price Index =", round(Fisher, 2), "\n")

cat("Marshall-Edgeworth Price Index =", round(ME, 2), "\n")
```

Home Work

Calculate the crime index for the year 2001 with 1991 as the base year if the data is reported in the format (crime in 1991, crime in 1992, weight): Robbery (13, 8, 6), Car Theft (15, 22, 5), Motorcycle Theft (249, 185, 4), Pocket Picking (328, 259, 1) and Theft by Servant (497, 448, 2).

TIME SERIES ANALYSIS IN R

A time series can be defined as a collection of observations made sequentially through time, especially equal intervals of time. Time series occur in various fields. Examples of time series include mean monthly atmospheric temperature data, quarterly gross domestic product data, sales figures in successive days, weeks, months, quarters or years, monthly birth totals in Abia state and so on. Thus, methods of analyzing time series constitute an important area of statistics. Time series analysis is an area of specialization in statistics that deals with the methods of analyzing and forecasting time series.

Two classifications of time series are worthy of mention. Discrete time series occur when observations are taken at specific times, especially equal intervals of time. The dividend paid to shareholders in successive years is a discrete time series. On the other hand, a time series is said to be continuous if observations are made continuously through time, irrespective of whether the variable under consideration is discrete or continuous. In fact, they are mechanical recorders that take measurements continuously and produce a continuous trace rather than observations at discrete intervals of time. For instance, in many laboratories, it is important to keep temperature and humidity as constant as possible. As a consequence, devices are installed to measure these variables continuously. Action may be taken when the trace goes outside pre-specified limits. Visual examination of the trace may be necessary in several situations. However, for more detailed analysis, it is customary to convert the continuous trace to a series in discrete time by sampling the trace at appropriate equal intervals of time.

The interesting feature of time series analysis is the fact that successive observations are usually not independent. Therefore, the analysis must take into account the time order of the observations. When successive observations are dependent, future values may be predicted from past observations. If a time series can be predicted exactly, it is said to be deterministic. Most time series are stochastic in nature. For such time series, the future is only partly determined by past values. This implies that exact predictions are impossible and must be replaced by the idea that future values have a probability distribution, which is conditioned by knowledge of past values.

Objectives of Time Series Analysis

Time series are usually collected and analyzed for several reasons. The objectives of time series analysis are as follows:

(a) Description: A crucial step in the analysis of a time series is to plot the observations to obtain the graphical representation of time series called the time plot. The time plot can help us to determine certain features of time series. In particular, the time plot is useful in detecting the presence of trend, seasonal variation, outliers, turning points and structural breaks in a time series. Knowledge of features of a time series is fundamental to the choice of an appropriate method of analyzing the time series.

(b) Explanation: Suppose we have observations based on two time dependent variables. It may be possible to explain the variation in one of the time series using the variation in the other time series. For instance, we may wish to examine the relationship between sales and advertising expenditure. For this purpose, we need to fit an appropriate model to the data. Although regression models are sometimes used to model the relationships between two or more time dependent variables, there are situations where other models are needed. An example of such models is the transfer function model.

(c) Prediction: Given an observed time series, one may want to predict the future values of the time series. This can be done with the help of a time series model.

(d) Control: Time series are sometimes collected or analyzed for the purpose of improving control over some physical or economic system. For example, if a time series is generated that measures the 'quality' of a manufacturing process, the aim of the analysis may be to keep the process operating at a 'high' level. Control problems are closely related to prediction in many situations. For instance, if one can predict that a manufacturing process is going to move off target, then an appropriate corrective action can be taken. There are different control procedures. In statistical quality control, the observations are plotted on control charts and the controller takes action as a result of studying the charts. A more complicated procedure is based on modelling the data and using the model to find an 'optimal' control strategy. In this approach, a stochastic model is fitted to the series, future values of the series are predicted, and then the input process variables are adjusted so as to keep the process on target.

<u>Components of Time Series</u>

Traditional methods of analyzing time series deal with the decomposition of the time series into various components. The four known components of time series are the secular trend (trend), seasonal variation, cyclical variation and irregular variation. They are essentially the sources of variation in time series

Trend

This is the long term pattern in a time series or long term change in the mean level of a time series. Here, long term is a relative word. Thus, the nature of data that are being subjected to trend analysis determines what should be referred to as a long term or long period of time. For example, if we are studying the sales figures of a firm for 2021 and 2022 and we observe that they were more sales in 2014 than in 2013, it is not proper to conclude that there is trend in the data. This is because the period is too short to conclude that sales are showing an increasing tendency. On the other hand, when we put a strong germicide into bacterial culture and count the number of organisms after each 10 seconds for 8 minutes, the resulting observations showing a general pattern have secular trend. The two common types of trend are the linear and nonlinear trends.

The presence of trend in time series can be attributed to factors such as population change, technological advancement and large scale shifts in consumer taste. In particular, government intervention, technological changes, discovery, mass production and exhaustion of natural resources are major causes of growth or decline in many economic time series. The study of trends is useful in describing historical data and projecting past patterns into the future. It also plays a key role in the estimation and elimination of the trend component of a time series so as to study other components of the time series. Each of the time series graphed in Figure 1 has trend.

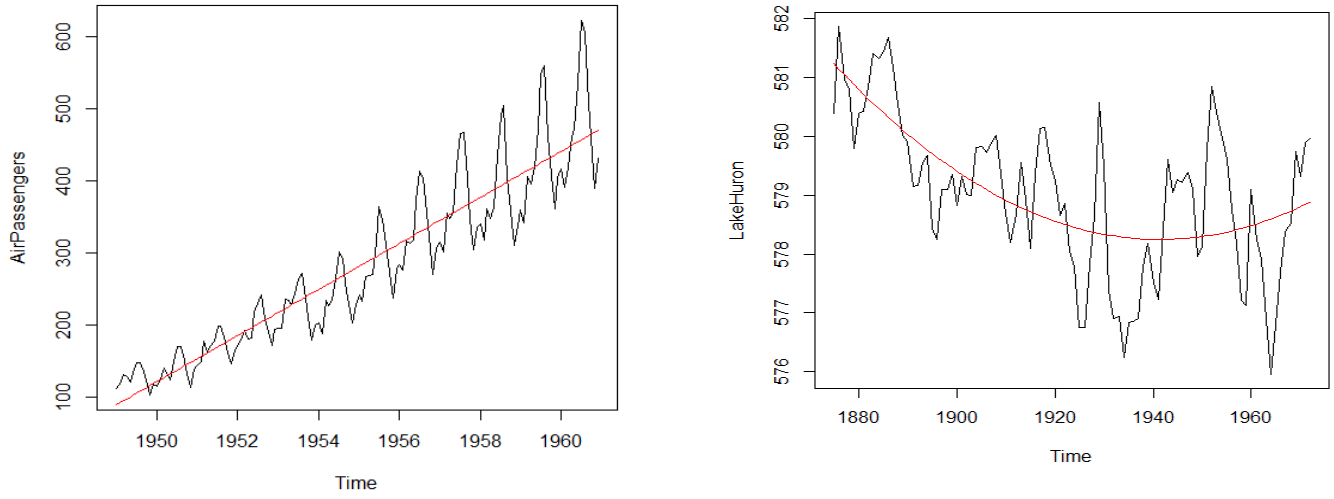


Figure 1: Time Plots of Data with Linear Trend (left panel) and Quadratic Trend (right panel)

Seasonal Variations

These are periodic fluctuations or movements in a time series that occur regularly every year. Seasonality is usually of a fixed and known frequency. Thus, seasonal variations can be predicted accurately. The air passenger data graphed in Figure 1 exhibit seasonal variations. Seasonal variations may be caused by factors such as weather conditions, custom and habit. For example, demand for umbrella is usually higher in rainy season than in dry season. Custom and habit are other factors among the factors that are responsible for seasonal variations in time series. There is a big demand for cash during festive periods. Most Christians buy chickens in the month of December for Christmas celebration.

The study and measurement of seasonal variations are a very critical aspect of time series analysis. Occasionally, knowledge of seasonal patterns is needed in production planning. There are situations where it is necessary to measure seasonal variation so as to study other types of variations (components) in time series. Seasonal indices are useful in scheduling purchases, inventory control, personnel requirements and advertising program.

Cyclical Variations

Cyclical variations are the long term oscillations or fluctuations about the trend. The duration of

the variations is usually longer than one year. These variations are neither regular in amplitude nor in length. A good number of time series pertaining to economics and business show cyclical variations. A typical business cycle comprises recurrent up and down movements of business activity about the trend (or normal). The following figure shows phases of business cycle.

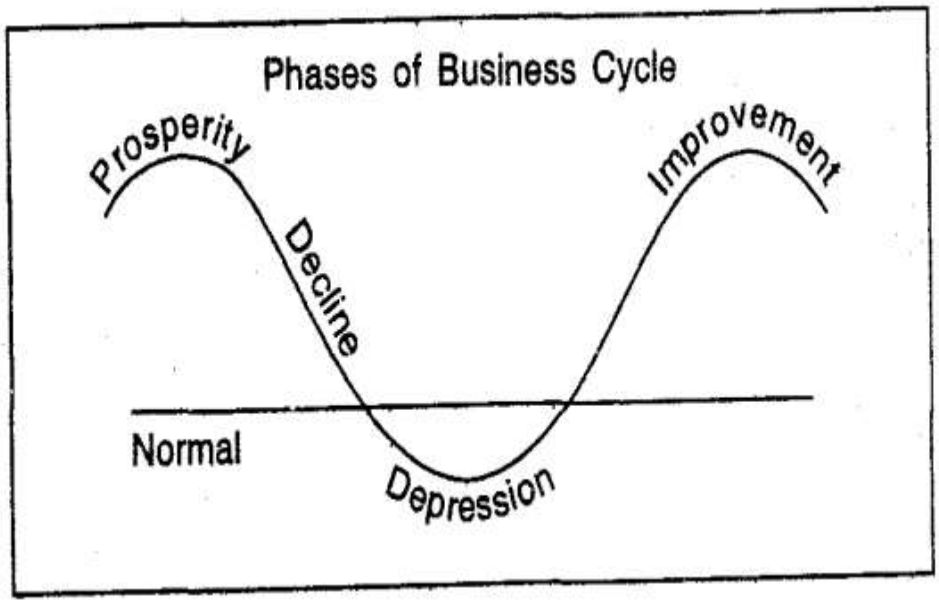


Figure 2: Phases of Business Cycle

The study of cyclical fluctuations is of utmost importance in formulating appropriate policies for stabilizing the level of business activity. Cyclical variations are the most difficult type of fluctuations to measure because of the following two reasons:

(a) Business cycles usually differ significantly in timing, amplitude and patterns, making their study quite difficult.

(b) Cyclical variations are mixed with the irregular variations. Thus, it may not be easy to isolate separately the cyclic and irregular variations.

Sometimes, people mistake cyclic variations for seasonal variations, but they are really different. The differences between seasonal and cyclical variations are as follows:

(a) If fluctuations are not of a fixed frequency, then they are cyclical. On the other hand, if the frequency is fixed and associated with some aspect of the calendar, then the pattern is seasonal.

(b) Seasonal variation takes place within a year whereas the duration of a cycle is at least two years.

(c) The magnitudes of cycles tend to be more variable than the magnitudes of seasonal patterns.

(d) A business cycle results from several causes, namely, the period of prosperity, decline (recession), depression and recovery (improvement). It is common to attribute seasonal variations in a particular time series to one cause.

Irregular Variations

The irregular variations or random variations are erratic variations that cannot be attributed to trend, seasonal variations and cyclic variations. These variations are caused by events such as floods, earthquakes, accidents, famines and strike actions. They are obviously unpredictable.

Classical Decomposition Models

Having discussed the various components of time series, it is worthwhile to explain models that are frequently used to decompose time series into their components. Here, we consider the three classical decomposition models. These are the additive model, multiplicative model and pseudo-additive (mixed additive-multiplicative) model.

The Additive Model

Let X_t be the value of a time series at time t. Suppose that the time series has all the four components such that its corresponding values of the trend, seasonal component, cyclical variation and irregular component at the time are T_t, S_t and I_t respectively. The additive (decomposition) model is

$$X_t = T_t + S_t + C_t + I_t \quad (1)$$

In the additive model, all the components act independently of one another. The model is suitable for the decomposition of a time series in which the amplitude of both seasonal and irregular variations do not change as the level of the trend rises or falls. Consider the time plot below.

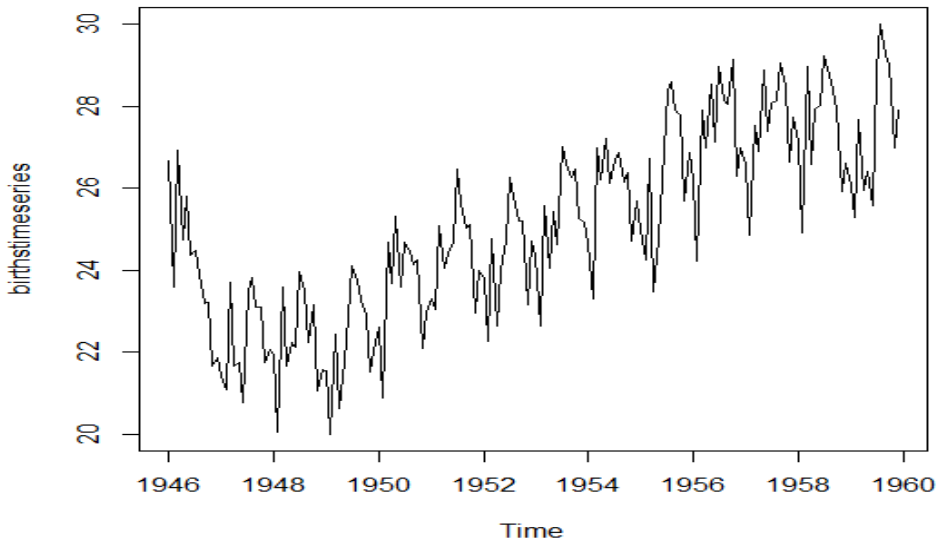


Figure 2: Time Series Plot of Number of Birth Per Month in New York City, from January, 1946 to December, 1959

The time series is seasonal and can be decomposed using the additive model because seasonal variations are relatively constant in size and do not depend on the level of the time series.

Multiplicative Model

In the multiplicative model, the value of a time series at time t is equal to the product of the values of the associated components of the time series. Thus, the model can be written as

$$X_t = T_t \times S_t \times C_t \times I_t \quad (2)$$

One of the assumptions of the model is that the components act proportionately to one another. In many time series, especially economic time series, the amplitude of the seasonal variation increases as the level of the time series increases. Whenever it is necessary to decompose the time series, the multiplicative model should be considered if none of the observed values of the series is zero. The figure below is the time plot of monthly sales in a souvenir shop at a beach resort town in Queensland, Australia, from January 1987 to December, 1993. The time series is a seasonal time series. It is obvious that seasonal variations vary with changes in the level of the time series. For this purpose, the multiplicative model can be used to decompose the time series.

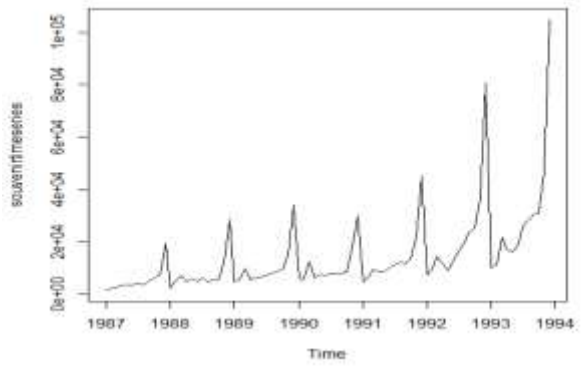


Figure 3: Time plot of Monthly Sales of Souvenirs

Interestingly, the log transformation can be used to convert the multiplicative model into the additive.

The Mixed Additive-Multiplicative Model

The multiplicative model is not suitable for analyzing a time series that contains very small or zero values. Such time series can be decomposed using the mixed additive-multiplicative model if the magnitude of the seasonal component depends on the trend. The mixed additive-multiplicative

model, pseudo-additive model or simply mixed model has some properties of both additive model and multiplicative model. In the model, the time series is expressed as a function of its components using both the addition and multiplication signs. One of the widely used mixed models is of the form

$$X_t = T_t \times S_t \times C_t + I_t.$$

(3)

Measurement of Trend

Several methods of measuring trend have been developed. They include the freehand method, method of semi-averages, moving average method and least squares method.

Freehand Method

In this method, we first of all obtain the time plot corresponding to the time series under consideration. A smooth freehand curve or line is then drawn through the plotted points in such a way that it represents the long term pattern in the time series. This becomes the trend line or curve for the time series, as the case may be.

Example 1: Fit a trend line to the data below using the freehand method

Year	2000	2001	2002	2003	2004	2005	2006	2007
Sales	30	46	25	59	40	60	38	65

Merits of the Freehand Method

(a) It is the simplest method of measuring trend
(b) It is applicable to both linear trend and nonlinear trend.

Demerits of the Freehand Method

(a) It is very subjective. Using this method, different people may fit different trend lines to the same data.
(b) Owing to the subjective nature of the method, predictions based on the method may not be reliable.

The Semi-Average Method

This method of estimating trend requires the splitting of the given data into two groups (halves) chronologically such that the groups contain the same number of observations. The arithmetic mean of observations belonging to each of the groups is computed and called a semi-average. Each of the averages is plotted against the center (middle) of the time frame it relates to. The straight line that passes through the points is the trend line obtained using the semi-average method.

Example 2: Use the semi-average method to fit a trend line to the following data:

Year	1971	1972	1973	1974	1975	1976	1977	1978
Production	40	45	40	42	46	52	56	61

Solution

The time series comprises 8 observations. If we split it chronologically into halves, then the first group contains the first four observations while the second group consists of the last four observations.

The mean of the first four observations $= \frac{40+45+40+42}{4} = 41.75.$

The arithmetic mean of the observations belonging to the second group $= \frac{46+52+56+61}{4} = 53.75.$

The middle of the time span corresponding to the average 41.75 is 1972.5 while that associated with the mean 53.75 is 1976.5. Therefore, we plot 41.75 and 53.75 against 1972.5 and 1976.5 respectively to obtain the trend line. Figure 4 contains the trend line and original time plot of the time series

Merits of the Semi-Average Method

(a) It is easy to apply the method.
(b) It is an objective method of measuring trend

Demerits of the Semi-Average Method

(a) It is not applicable to nonlinear trend.
(b) If there are extreme values in at least one half of the time series, the trend line obtained using this method will not give a true picture of the long term pattern in the series.

Remarks

Let n be a positive integer. If a time series that contains 2n+1 observations (i.e the total number of observations is an odd number) is arranged in chronological order, one half of the series contains the first n observations while the half contains the last n observations. In this (n+1)th observation is not a member of any of the halves. In the case of an even number of observations (say 2n), the first n observations belong to one half whereas the last n observations belong to the other half.

Moving average Method

The moving average of a period m is a series of successive arithmetic means of m observations at a time. Here, the first average is the mean of the first m observations. The second average is the mean of the m observations, starting from the second observation up to (m + 1)th observation. Similarly, the third average is the mean of the m observations, which are the third, fourth, … and (m + 2)th observations and so on. The computation of the means ends whenever it is impossible to obtain

the average of m observations on the given time series. Consider an annual time series comprising the observations $x_1, x_2, x_3, ..., x_n$. For the 3-year moving average, m=3 and we have the following averages:

$$\frac{x_1+x_2+x_3}{3}, \frac{x_2+x_3+x_4}{3}, \frac{x_3+x_4+x_5}{3}, \ldots$$

Suppose that m is odd. That m is of the form 2k+1, where k is a positive integer. Then each of the computed means is placed against the mid-value of the time span it corresponds to and the averages (means) are the trend values. If m is even, (i.e m=2k), each of the means is placed in between the two middle values of the corresponding time interval. In this case, the centered moving average is needed. The centered moving average the series of the arithmetic means of two of the computed averages at time. When m is even, the centered averages become the trend values since they correspond to periods in the original data.

Example 3: Calculate the 3-year moving averages of the data below:

Year	1941	1942	1943	1944	1945	1946	1947	1948	1949	1950	1951
Production(in million tonnes)	225	210	201	215	223	245	235	225	233	249	265

Solution

Year	Production	3-Yearly Moving Total	3-Yearly Moving Average (col 3/3)
1941	225	-	-
1942	210	636	212
1943	201	626	208.67
1944	215	639	213
1945	223	683	227.67
1946	245	703	234.33
1947	235	713	237.67
1948	225	693	231
1949	233	707	235.67

1950	249	747	249
1951	265	-	-

Merits of the Moving Average Method

(a) This method is easy to understand and apply.

(b) It is an objective method.

(c) It is a flexible method because if few observations are added, the entire calculations are not changed.

(d) If the period of oscillatory movements is equal to the period of the moving average, these movements are completely eliminated.

Demerits of the Moving Average Method

(a) It is not possible to calculate trend values for all the items of the series. Some information is always lost at its ends.

(b) If the appropriate trend model for the given data is not at least approximately linear and the oscillatory and the random fluctuations are not uniform in terms of period and amplitude, this method will not give correct values of the trend.

(c) The trend values obtained by moving averages may not follow any mathematical pattern. Thus, the moving average method cannot be used for forecasting future trend values.

(d) Like arithmetic mean, moving averages are affected by extreme values.

The Least Squares Method

This is a widely used method of fitting a trend line or curve. It is aimed at obtaining the best fitting trend line or curve by minimizing the sum of squares of the deviations of the series from the trend component. In this method, the trend component (T_t). In what follows, the process of fitting each of a straight line trend model, quadratic trend model and exponential trend model by least squares method is explained.

Fitting a Straight Line Trend by Least Squares Method

Here, $T_t = \sum_{i=0}^{1} a_i t^i = a_0 + a_1 t$. Therefore, the sum of squares of the deviations of X_t from T_t is $Q = \sum (X_t - a_0 - a_1 t)^2$. Finding and equating the partial derivative of Q with respect to each

of a_0 and a_1 to zero and replacing a_0 and a_1 by their estimators $\hat{a}_0$ and $\hat{a}_1$ respectively, we obtain the following normal equations:

$$\sum X_t = n\hat{a}_0 + \hat{a}_1 \sum t \tag{6}$$

$$\sum tX_t = \hat{a}_0 \sum t + \hat{a}_1 \sum t^2 \tag{7}$$

In (6), n is the total number of observations that make up the time series. Values of t can be chosen based on any time origin. For simplicity, we shall consider the midpoint in time as the origin. When n is odd, the values of the time variable t are obtained by subtracting the median of the time points (periods) for which data are available from each of the period. When n is even, each of the deviations is multiplied by 2. In both cases, $\sum t = 0$ and we have

$$\sum X_t = n\hat{a}_0 . \tag{8}$$

$$\sum tX_t = \hat{a}_1 \sum t^2 \tag{9}$$

From (8), we have

$$\hat{a}_0 = \frac{\sum X_t}{n} = \overline{X}.$$

Using (9), we obtain

$$\hat{a}_1 = \frac{\sum tX_t}{\sum t^2} .$$

Example 4: Using the least squares method, fit a straight line trend to the data in Example 3.

Solution: Here, n=11. Hence, t=Year-1946. Thus, we obtain the results in the table below.

Year	X_t	t=Year-1946	tX_t	t^2
1941	225	-5	-1125	25
1942	210	-4	-840	16
1943	201	-3	-603	9
1944	215	-2	-430	4
1945	223	-1	-223	1
1946	245	0	0	0
1947	235	1	235	1
1948	225	2	450	4
1949	233	3	699	9
1950	249	4	996	16
1951	265	5	1325	25
Total	2526	0	484	110

The least estimates of the parameters are calculated as follows

$$\hat{a}_0 = \frac{\sum X_t}{n} = \frac{2526}{11} = 229.64.$$

$$\hat{a}_1 = \frac{\sum tX_t}{\sum t^2} = \frac{484}{11} = 4.40.$$

The estimated linear trend equation is

$$\hat{T}_t = 229.64 + 4.4t.$$

Fitting a Quadratic Trend Model by Method of Least Squares

Consider the quadratic (parabolic) trend equation

$$T_t = \sum_{i=0}^{2} a_i t^i = a_0 + a_1 t + a_2 t^2.$$

The least estimates of $\hat{a}_0$, $\hat{a}_1$ and $\hat{a}_2$ of a_0, a_1 and a_2 respectively, are determined through the minimization of $Q = \sum (X_t - a_0 - a_1 t - a_2 t^2)^2$. If we obtain the values of t by taking deviations about the median of the given periods, then we need to solve the normal equations below to obtain the estimates:

$$\sum X_t = n\hat{a}_0 + \hat{a}_2 \sum t^2 \tag{10}$$

$$\sum tX_t = \hat{a}_1 \sum t^2 \tag{11}$$

$$\sum t^2 X_t = \hat{a}_0 \sum t^2 + \hat{a}_2 \sum t^4 \tag{12}$$

From (10),

$$\hat{a}_2 = \frac{\sum X_t - n\hat{a}_0}{\sum t^2}.$$

Thus, Equation (12) can be written as

$$\sum t^2 X_t = \hat{a}_0 \sum t^2 + \frac{\sum X_t - n\hat{a}_0}{\sum t^2} \sum t^4.$$

This implies that

$$\sum t^2 X_t \sum t^2 = \hat{a}_0 \left(\sum t^2\right)^2 + \sum X_t \sum t^4 - n\hat{a}_0 \sum t^4.$$

If we rearrange the equation above and solve for $\hat{a}_0$, we have

$$\hat{a}_0 = \frac{\sum X_t \sum t^4 - \sum t^2 X_t \sum t^2}{n\sum t^4 - \left(\sum t^2\right)^2}.$$

Again,

$$\hat{a}_2 = \frac{\sum X_t - n\hat{a}_0}{\sum t^2} = \frac{n\sum t^2 X_t - \sum X_t \sum t^2}{n\sum t^4 - \left(\sum t^2\right)^2}.$$

Based on Equation (11),

$$\hat{a}_1 = \frac{\sum tX_t}{\sum t^2}.$$

Example 5: Fit a parabolic trend to the following time series data and estimate the production in 2019:

Year	2008	2009	2010	2011	2012	2013	2014
Production (in '000' units)	42	49	62	75	92	122	158

Solution

Year	X_t	t=Year-2011	t^2	t^3	t^4	$t\ X_t$	$t^2 X_t$
2008	42	-3	9	27	81	-126	378
2009	49	-2	4	8	16	-98	196
2010	62	-1	1	1	1	-62	62
2011	75	0	0	0	0	0	0
2012	92	1	1	1	1	92	92
2013	122	2	4	8	16	244	488
2014	158	3	9	27	81	474	1422
Total	600	0	28	0	196	524	2638

For the estimates, we have

$$\hat{a}_0 = \frac{\sum X_t \sum t^4 - \sum t^2 X_t \sum t^2}{n\sum t^4 - \left(\sum t^2\right)^2} = \frac{600\times 196 - 2638\times 28}{7\times 196 - 28^2} = \frac{43736}{588} = 74.38;$$

$$\hat{a}_1 = \frac{\sum tX_t}{\sum t^2} = \frac{524}{28} = 18.71;$$

$$\hat{a}_2 = \frac{n\sum t^2 X_t - \sum X_t \sum t^2}{n\sum t^4 - \left(\sum t^2\right)^2} = \frac{7\times 2638 - 600\times 28}{588} = \frac{1666}{588} = 2.83.$$

Therefore, the estimated parabolic trend equation is

$$\hat{T}_t = 74.38 + 18.71t + 2.83t^2.$$

For 2019, t=8. Hence, the estimate of production in 2019 is given by

$$\hat{T}_9 = 74.38 + 18.71 \times 8 + 2.83 \times 8^2 = 405.18 \text{ thousand units.}$$

Exponential Trend

If a given time series increases or decreases at a constant rate without approaching some upper limit, the long term pattern in the series can be represented by an exponential curve. The formula for the exponential curve is

$$X_t = ab^t.$$

Equivalently,

$$\log X_t = \log a + t \log b.$$

The values of a and b are obtained by solving the normal equations:

$$\sum \log X_t = n \log a + \log b \sum t \quad (13)$$

$$\sum (t \log X_t) = \log a \sum t + \log b \sum t^2 \quad (14)$$

When deviations are taken from the middle of the period to which the time series corresponds, Equations (13) and (14) take the following form:

$$\sum \log X_t = n \log a \Rightarrow a = \text{antilog}\left(\frac{\sum \log X_t}{n}\right);$$

$$\sum (t \log X_t) = \log b \sum t^2 \Rightarrow b = \text{antilog}\left(\frac{\sum (t \log X_t)}{\sum t^2}.\right)$$

Example 5: The sales of a company for the years 2003 to 2009 are given below

Year	2003	2004	2005	2006	2007	2008	2009
Sales (in millions)	32	47	65	92	190	132	275

Find the trend values by using the equation $X_t = ab^t$ and estimate the sales for 2012.

Solution

Year	X_t	T	$\log X_t$	t^2	$t \log X_t$	Trend Value= $\text{antilog}(1.9704 + 0.149 \times t)$

2003	32	-3	1.5051	9	-4.5153	33.3734
2004	47	-2	1.6721	4	-3.3442	47.0327
2005	65	-1	1.8129	1	-1.8129	66.2827
2005	92	0	1.9638	0	0	93.4114
2007	190	1	2.2788	1	2.2788	131.6437
2008	132	2	2.1206	4	4.2412	185.5240
2009	275	3	2.4393	9	7.3179	261.4568
Total		0	13.7926	28	4.1655	

$$\log a = \frac{\sum \log X_t}{n} = \frac{13.7926}{7} = 1.9704.$$

$$\log b = \frac{\sum (t \log X_t)}{\sum t^2} = \frac{4.1655}{28} = 0.149.$$

Thus,

$$\log X_t = 1.9704 + 0.149t.$$

For 2012, t=9. Hence, the estimated sales in 2012 is

$$X_6 = \text{antilog}(1.9704 + 0.149 \times 6) = 731.8.$$

Merits of the Least squares Method

(a) Given the mathematical form of the trend to be fitted, the least squares method is an objective method.

(b) Unlike the moving average method, it is possible to compute trend values for all the periods and predict the value for a period lying outside the observed data.

(c) The results of the method of least squares are most satisfactory because the fitted trend satisfies the two most important properties, namely, $\sum\left(X_t - \hat{T}_t\right) = 0$ and $\sum\left(X_t - \hat{T}_t\right)^2$ is minimum.

(d) It can be used to estimates parameters of both linear and nonlinear trend equations.

Demerits of the Least squares Method

(a) As compared with the moving average method, it is a cumbersome method.

(b) It is not flexible like the moving average method. If some observations are added, then the entire calculations are to be done once again.

(c) It cannot be used to fit growth curves.

Growth Curves

Several growth curves have been used to measure trend. The two commonly used growth curves are the Gompertz curve and logistic curve. The equation of the Gompertz curve is

$$X_t = Ka^{b^t}.$$

For the logistic curve, we have

$$\frac{1}{X_t} = K + ab^t.$$

Changing the Trend Origin

Although the computation of trend values is simplified by considering the origin to be the median of the years to which the data correspond, there are situations where it is necessary to change the origin of the trend equation. For the linear trend equation, the formula for calculating the trend values when the origin has been shifted from t to $t + k$ is

$$\hat{T}_{t+k} = \hat{a}_0 + \hat{a}_1(t + k).$$

Similarly, for the quadratic trend equation, we obtain

$$\hat{T}_{t+k} = \hat{a}_0 + \hat{a}_1(t + k) + \hat{a}_2(t + k)^2.$$

Conversion of Annual Trend Values to Monthly Trend Values

Consider a linear trend equation based on annual totals. The annual trend equation is converted into a trend equation for monthly totals using the formula:

$$\hat{T}_t = \frac{\hat{a}_0}{12} + \frac{\hat{a}_1}{144}t.$$

If the quadratic trend equation is applicable, the conversion from a trend equation based on annual totals to a trend equation based on monthly totals is done using the equation:

$$\hat{T}_t = \frac{\hat{a}_0}{12} + \frac{\hat{a}_1}{144}t + \frac{\hat{a}_1}{1728}t^2.$$

Measurement of Seasonal Variations

Many time series exhibit seasonal patterns. However, annual time series cannot be seasonal. That is annual time series do exhibit seasonal movements. The four common methods of measuring seasonal variations include the method of simple averages, ratio to trend method, ratio to moving average method and link relative method.

Method of Simple Averages

This is the simplest method of finding a seasonal index. Based on this method, we can calculate seasonal indices for quarterly time series data by taking the following steps:

(i) Arrange the original data to years and quarters.

(ii) Find the quarterly totals (i. e. sum the observations pertaining to each quarter over the years).

(iii) Calculate the quarterly means (i. e divide each quarterly total by the number of years for which the data are available).

(iv) Determine the average of the quarterly means by dividing the sum of the quarterly averages by 4.

(v) Calculate the seasonal indices using the formula

$$\text{Seasonal Index for jth Quarter} = \frac{\left(\begin{array}{l}\text{Average of Observations}\\ \text{Corresponding to jth Quarter}\end{array}\right)}{\text{Average of Quarterly Average}} \times 100,\ j = 1, 2, 3, 4.$$

The steps above are also necessary for the computation of seasonal index based on monthly time series except that quarters are replaced by months.

Example 6: Use the simple average method to compute seasonal indices for the data in the following table:

	Quarter			
Year	Q_1	Q_2	Q_3	Q_4
2001	79	105	95	89
2002	89	137	132	116
2003	105	153	142	127
2004	142	201	179	164
2005	238	243	230	216

Solution

	Quarter			
Year	Q_1	Q_2	Q_3	Q_4
2001	79	105	95	89
2002	89	137	132	116
2003	105	153	142	127
2004	142	201	179	164
2005	238	243	230	216
Quarterly Average	130.6	167.8	155.6	142.4

Sum of the quarterly averages=130.6=167.8+155.6+142.4=596.4.

Average of the quarterly averages $= \frac{596.4}{4} = 149.1$.

Thus,

Seasonal index for the first quarter = $\frac{130.6}{149.1} \times 100 = 87.6.$

Seasonal index for the second quarter = $\frac{167.8}{149.1} \times 100 = 112.5.$

Seasonal index for the third quarter = $\frac{155.6}{149.1} \times 100 = 104.4.$

Seasonal index for the fourth quarter = $\frac{142.4}{149.1} \times 100 = 95.5.$

It is easy to verify that the sum of the indices is 400. Notably, the seasonal indices ought to sum up to 400 when quarterly time series data are used. If monthly data are considered, the indices should add up to 1200. When the required sum is not obtained, the indices have to be adjusted. For quarterly data, the adjustment requires the multiplication of each of the computed indices by 400 and division of the product by the sum of the indices. In the case of the monthly data, each index is multiplied by 1200 and the product is divided by sum of the calculated indices to obtain the adjusted seasonal indices.

Merits of the Method of Simple Averages

(i) It is easy to apply.

(ii) It is an objective method.

Demerit of the Method of Simple Averages

(i) It is not a good method of obtaining seasonal indices from a time series that has trend.

Ratio-to-Trend Method

The ratio-trend method is also called the percentage-to-trend method. It is an improvement over the method of simple averages. In this method, seasonal variation for a given month (or quarter) is assumed to be a constant fraction of trend. Consider a quarterly time series. The following steps are necessary for calculating seasonal indices for the data using this method:

(a) Compute the trend values by applying the least squares method.

(b) Multiply each of the given values of the time series by 100 and divide the product by the corresponding trend value. The essence of taking this step is to detrend the time series by virtue of the multiplicative model.

(c) Isolate the cyclical and irregular variations from the figures in Step b by finding quarterly averages of the figures. Quarterly means should be considered if the figures contain no extreme value. If the figures comprise at least one extreme, quarterly medians are preferable to quarterly means.

(d) Calculate seasonal indices from the averages in Step c by multiplying each of them by 100 and dividing the product by the mean of the averages. Notably, the figures obtained in this step are reported as seasonal indices if their sum is 400. Whenever the sum is not approximately equal to 400, the results are adjusted to obtain the requisite seasonal indices. To adjust each of the figures obtained in Step d, we multiply it by 400 and divide the product by sum of the figures.

The ratio-to-trend can also be used to calculate seasonal indices from monthly time series data by applying the steps above and replacing quarterly averages by monthly averages. For monthly data, the sum of the indices should be 1200. Otherwise, we adjust each index by multiplying it by 1200 and dividing the product by sum of the indices.

Example 7: Using the ratio-to-trend method, find seasonal indices for the following data:

	Quarter			
Year	Q_1	Q_2	Q_3	Q_4

2003	30	40	36	34
2004	34	52	50	44
2005	40	58	54	48
2006	54	76	68	62
2007	80	92	86	82

Solution

The following table contains the annual totals and means.

	Quarter					
Year	Q_1	Q_2	Q_3	Q_4	Annual Total	Annual Mean
2003	30	40	36	34	140	35
2004	34	52	50	44	180	45
2005	40	58	54	48	200	50
2006	54	76	68	62	260	65
2007	80	92	86	82	340	85

To obtain the estimated annual trend equation, we consider the information in the table below.

Year	Annual Mean (x)	t	tx	t^2	Annual Trend Values
2003	35	-2	-70	4	32
2004	45	-1	-45	1	44
2005	50	0	0	0	56
2006	65	1	65	1	68
2007	85	2	170	4	80
Total	280	0	120	10	

Consequently,

$$\hat{a}_0 = \bar{x} = \frac{\sum x}{n} = \frac{280}{5} = 56.$$

$$\hat{a}_1 = \frac{\sum tx}{\sum t^2} = \frac{120}{10} = 12.$$

The estimated equation of the straight line trend is $\hat{T}_t = \hat{a}_1 + \hat{a}_1 t = 56 + 12t$.

Hence, the quarterly increment is $\frac{\hat{a}_1}{4} = \frac{12}{4} = 3.$

Calculation of Quarterly Trend Values

Consider 2003. The trend value for the middle quarter (i.e half of the second and third quarters) is 32. Since the quarterly increment is 3, the trend values for the second quarter is $32 - \frac{3}{2} = 30.5$ while the trend value for the third quarter of the year is $32 + \frac{3}{2} = 33.5.$ The trend value for the first quarter of the same year is $30.5 - 3 = 27.5.$ For the last quarter of the year, we have the trend value $33.5 + 3 = 36.5$ and so on. The table below contains the quarterly trend values:

Year	Q_1	Q_2	Q_3	Q_4
2003	27.5	30.5	33.5	36.5
2004	39.5	42.5	45.5	48.5
2005	51.5	54.5	57.5	60.5
2006	63.5	66.5	69.5	72.5
2007	75.5	78.5	81.5	84.5

Expressing the given observations as percentages of their corresponding trend values, we obtain the following table:

Year	Q_1	Q_2	Q_3	Q_4
2003	109.09	131.15	107.46	93.15
2004	86.08	122.35	109.89	90.72
2005	77.67	106.42	93.91	79.34
2006	85.04	114.29	97.84	85.52
2007	105.96	117.20	105.52	97.04
Total	463.84	591.41	514.62	445.77
Average (Mean)	92.77	118.28	102.92	89.15
SI	92.05	117.36	102.12	88.46

Sum of the averages $= 92.77 + 118.28 + 102.92 + 89.15 = 403.12.$

The seasonal indices are computed below

Seasonal index (SI) for the first quarter $= \frac{92.77 \times 400}{403.12} = 92.05.$

SI for the second quarter $= \frac{118.28 \times 400}{403.12} = 117.36.$

SI for the third quarter $= \frac{102.92 \times 400}{403.12} = 102.12.$

SI for the fourth quarter $= \frac{89.15 \times 400}{403.12} = 88.46.$

Merit of Ratio-to-Trend Method

(a) It is easy to understand.

(b) It is a better method of measuring seasonal variations than the method of simple averages.

(c) It is an objective method.

Limitation of Ratio-to-Trend Method

(a) It is more difficult to apply this method than the method of simple averages.

(b) Seasonal indices computed by ratio-to-trend may be unreliable when there are very noticeable cyclical movements in the time series.

Ratio-to-Moving-Average Method

The ratio-to-moving-average method, also known as the percentage of moving average method, is arguably the most widely used method of computing seasonal indices. For monthly time series data, this method is based on the following steps:

(a) Compute the centered 12-month moving average (MA). Note that if monthly data are under consideration, the centered 4-quarter moving average must be used.

(b) Express each of the observed value of the series as a percentage of the corresponding centered 12-month moving average.

(c) Find the monthly averages. The averages are the same as the monthly means if there are no extreme values among the figures computed in Step b. Otherwise, monthly medians are computed.

(d) Multiply of each of the monthly averages by 100 and divide the sum by the average of the monthly averages. The figures obtained in this step are seasonal indices if their sum is

equal to 1200. When their sum is not 1200 for monthly data, they need to be adjusted. On the other, if seasonal indices are from quarterly data based on this method, they should some up to 400. If their sum is not 400. An adjustment is necessary.

Example 8: Calculate seasonal indices using ratio-to-moving-average method.

	Quarter			
Year	Q_1	Q_2	Q_3	Q_4
2005	68	62	61	63
2006	65	58	66	61
2007	68	63	63	67

Solution

Year	Quarter	X_t	4-Quarter Moving Total	Centered 4-Quarter Moving Avearge (i.e 2-Quarter Moving Total of column $4 \div 8$)	Given Figure as % of Centered 4-Quarter MA (i.e $\frac{\text{Column } 3 \times 100}{\text{Column } 5}$)
2005	I	68			
	II	62			
			254		
	III	61		63.125	96.63
			251		
	IV	63		62.250	101.20
			247		
	I	65		62.375	104.21
			252		
	II	58		62.750	92.43
			250		

	III	66		62.875	104.97
			253		
	IV	61		63.875	95.50
			258		
	I	68		64.125	106.04
			255		
	II	63		64.500	97.67
			261		
	III	63			
	IV	67			

Arranging the figures in the last column of the table above (percentage to moving average) according to years and quarters, we obtain the following table:

Year	Quarter			
	Q_1	Q_2	Q_3	Q_4
2005	-	-	96.63	101.20
2006	104.21	92.43	104.97	95.50
2007	106.04	97.67	-	-
Total	210.25	190.10	201.60	196.70
Average	105.125	95.05	100.80	98.35

Here, the arithmetic mean of the quarterly averages is

$$\frac{105.125+95.05+100.80+98.35}{4}=\frac{399.32}{4}=99.83.$$

The seasonal indices are calculated below:

Seasonal index for the first quarter $=\frac{105.125\times100}{99.83}=105.30.$

Seasonal index for the second quarter $=\frac{95.05\times100}{99.83}=95.21.$

Seasonal index for the third quarter $= \frac{100.80 \times 100}{99.83} = 100.97.$

Seasonal index for the fourth quarter $= \frac{98.35 \times 100}{99.83} = 98.52.$

Merits of the Moving Average Method

(i) Seasonal indices that are obtained using this method are usually more representative of the given data than the indices based on the same data and ratio-to-trend method.

(ii) The method is easy to understand.

(iii) It is flexible.

Demerits of the Moving Average Method

Since this method employs the moving average method to obtain the trend values, the trend values corresponding to the first few observations and equal number of observations in the end are not available.

Link Relative method

This method of measuring seasonal variations comprises the following steps:

(i) Compute the requisite link relatives. The link relative for a particular period is obtained by expressing the observed value of the series in the period as a percentage of the value of the series in the previous period. Hence, the link relative for the ith period is

$$LR_i = \frac{\text{value of the time series in ith period}}{\text{value of the time series in (i-1)th period}} \times 100, i = 2, 3, ..., n.$$

(ii) Calculate average of the link relatives for each season (month or quarter). If there is no case of extreme values, the arithmetic mean of the link relatives for each season is appropriate. Otherwise, the median should be considered.

(iii) Convert the average link relatives in Step ii into chain relatives on the basis of the first season. Here, we assume that the chain relative for the first season is 100. For clarity, the assumed chain relative for the first season is denoted by ACR_1. The chain relative for the jth season (CR_j) is mathematically defined as

$$CR_j = \frac{\left(\begin{array}{l}\text{average link relative} \\ \text{for jth season}\end{array}\right) \times \left(\begin{array}{l}\text{chain relative} \\ \text{for (j-1)th season}\end{array}\right)}{100}, j = 2, 3, ..., S,$$

where S=12 for monthly data or 4 for quarterly data.

(iv) Calculate the chain relative for the first season based on that of the season. Let the calculated chain relative be CCR_1. Then,

$$CCR_1 = \frac{\begin{pmatrix}\text{average link relative}\\ \text{for the first season}\end{pmatrix}\times\begin{pmatrix}\text{chain relative}\\ \text{for the last season}\end{pmatrix}}{100}.$$

(v) Compute $C = \frac{CCR_1 - 100}{S}$. The essence of taking this step is to enable us obtain the adjusted chain relatives since CCR_1 is usually not equal to 100 (ACR_1).

(vi) Determine the adjusted chain relatives using the formula
$ACR_j = CR_j - (j-1)C$, j=2,3,..., S.

(vi) Divide the sum of 100 and the adjusted chain relatives by S.

(vii) Express each of the figures in the numerator of the fraction in Step vi as a percentage of the average of the figures to obtain the seasonal index for the corresponding season. Note that the first figure is used to obtain the seasonal index for the first season while ACR_j (j=2,3,..., S) in Step vi are used for the computation of the seasonal indices for the remaining (S-1) seasons.

Example 9: Apply the link relative method to the data below and calculate seasonal indices:

	Quarter			
Year	Q_1	Q_2	Q_3	Q_4
2001	80.40	69.68	58.96	83.08
2002	93.80	75.04	58.96	96.48
2003	83.08	77.72	72.36	85.76
2004	91.12	83.08	67.00	93.80
2005	91.20	58.96	53.60	88.44

Solution

Link relative for the observation in the second quarter of 2001 = $\frac{69.68}{80.40}$ x 100 = 86.67

Link relative for the observation in the third quarter of 2001 = $\frac{58.96}{69.68}$ x 100 = 84.62

etc.

The link relatives are contained in the following table:

	Quarterly link relatives			
YEAR	**Q1**	**Q2**	**Q3**	**Q4**
2001		86.67	84.62	140.91
2002	112.9	80	78.57	163.64
2003	86.11	93.55	93.1	118.52

2004	106.25	91.18	80.65	140
2005	97.23	64.65	90.91	165
Total	**402.49**	**416.05**	**427.85**	**728.07**
Average	**100.62**	**83.21**	**85.57**	**145.61**

Assumed value of the chain relative for the first quarter = 100

Chain relative for the second quarter = $\frac{\text{Average of link relative for the second quarter}}{100}$ x 100

$= \frac{83.22}{100}$ x 100 = 83.22

Chain relative for the third quarter = $\frac{85.57 \text{ x } 83.22}{100}$ = 71.21

Chain relative for the fourth quarter = $\frac{145.61 \text{ x } 71.21}{100}$ = 103.69

New chain relative for first quarter on the basis of the last quarter = $\frac{100.60 \text{ x } 103.69}{100}$ = 104.30

C = $\frac{1}{4}$ [104.30 - 100] = $\frac{4.50}{4}$ = 1.08

Adjusted chain relative for second quarter = 83.22 – (2-1) x 1.08 = 82.14

Similarly,

Adjusted chain relative for third quarter = 71.21 – (3-1) x 1.08 = 69.05

and

Adjusted chain relative for fourth quarter = 103.69 – 3 x 1.08 = 100.45

Average of adjusted chain relative = $\frac{100+82.14+69.05+100.45}{4}$ = 87.91

Seasoned index for ith quarter = $\frac{\text{Adjusted relative for the ith quarter}}{\text{Average of the adjusted chain relative}}$ x 100

Therefore, seasonal index for the first quarter = $\frac{100 \text{ X } 100}{87.91}$ = 113.75

seasonal index for the second quarter = $\frac{82.14 \text{ X } 100}{87.91}$ = 93.44

seasonal index for the third quarter $= \frac{69.05 \text{ X } 100}{87.91} = 78.55$

seasonal index for the fourth quarter $= \frac{100.45 \text{ X } 100}{87.91} = 114.26$

Note that the sum of the indices = 113.75 + 93.44 + 78.55 +114.26 = 400

Measurement of Cyclical Variations by the Residual method

The cyclical variations are often measured indirectly as a residue after the elimination of other components of a time series. In this regard, we take the following steps;

1. Compute the trend values (by least squares method) and the seasonal indices by appropriate methods (preferably ratio moving average method). Here, the seasonal indices are obtained as fractions rather than ordinary percentages.
2. Assuming multiplicative method, divide each value of the observed values of time series by the product of the corresponding trend value and seasonal index. This ratio comprises the cyclical and random components.

   That is, C. R $= \frac{X_t}{T_{tSt}}$

3. Compute an appropriate moving average (say of order 3) of the values obtained in step 2.

Other methods of measuring cyclical variation include; Reference cycle analysis, Direct percentage variation method and harmonic analysis.

Measurement of Irregular variations

Like the cyclical variations, the irregular component can also be obtained as a residue after eliminating the effects of other components.